

Steve Cabrel Tegua Kouam
Department of Physics, Faculty of Science
University of Douala, Cameroon
`steve.tegua@facsciences-uy1.cm`

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Editorial Office
Applied Energy
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Dear Editor,

We are pleased to submit our manuscript entitled “**Quantum Agrivoltaic Digital Twin: Coordinated Vibronic Light-Harvesting, Environmental Calibration, and Agro-Rural Cybersecurity**” for consideration as a Research Article in *Applied Energy*. It is not under consideration elsewhere; the journal’s systems-level energy–agriculture–water scope is its natural home.

Significance of the work

Standard semi-transparent solar panels act as passive spectral filters in agrivoltaics. Using numerically exact quantum dynamics (PT-HOPS/SBD), we show that an organic photovoltaic layer integrated with a nanoparticle-on-mirror (NPoM) plasmonic cavity can *actively* shape the transmitted spectrum to protect quantum coherence in the photosynthetic machinery, while the same cavity enables label-free SERS diagnostics of crop health. The digital twin links these quantum effects to canopy yields, Net Ecological Benefit, and cooperative economics.

Key advances

- **Coherence protection:** dual-band filtering at 750 nm and 820 nm extends exciton coherence by 50 % and raises the forward transfer yield by 25 % at room temperature (100 disorder realizations).
- **Experimental testability:** all modelled observables are anchored to independent experimental benchmarks, with two validation protocols (dual-band 2DES with SLM pulse shaping; time-resolved NPoM SERS).
- **SERS–transport trade-off resolved:** the NPoM diverts 79–84 % of the excitation to the plasmon mode, but confining it to a 1 % sentinel fraction preserves 97.1 % of the global canopy yield.
- **Farm-scale proof of concept:** a 500 m² cooperative greenhouse in Cameroon combines 90 MWh yr^{−1} of OPV electricity, 28 % evapotranspiration savings, a Net Ecological Benefit of 19.6 kg (CO₂)e/m²/yr, and a 4.23 yr payback under 30 % subsidy, with BB84-secured (4.8 % QBER) IoT telemetry.

Why *Applied Energy*?

The journal’s readership spans photovoltaics, agricultural-energy systems, the water–energy–food nexus, lifecycle assessment, and energy economics — precisely the scales this work integrates, with all predictions falsifiable through the proposed protocols.

Novelty, compliance, and declarations. The integration of spectral coherence protection, NPoM SERS diagnostics, carbon-financed cooperative economics, and QKD-secured IoT in one platform is, to our knowledge, unreported. The manuscript has been checked by a native English speaker with expertise in the field; all authors approved it, declare no competing interests, and are available to review at least three other *Applied Energy* articles this year. An earlier version was considered at *Nature Energy*; we submit directly here as the more suitable venue. Subscription publication (no APC due); the corresponding author is based in Cameroon.

Suggested reviewers: A. Olaya-Castro (UCL), J. Yuen-Zhou (UCSD), J. J. Baumberg (Cambridge), G. A. Barron-Gafford (Arizona) — covering quantum biology, polaritonic chemistry, NPoM/SERS, and agrivoltaics.

Sincerely,

Steve Cabrel Tegua Kouam
(On behalf of all authors)

Enclosures: Manuscript, Supplementary Material (SI.pdf), Highlights, Graphical Abstract.